

Assignment No 5: Take appropriate dataset and apply

- ✓ AdaBoost algorithm to demonstrate that successive model enhances performance by progressively reducing bias.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.ensemble import AdaBoostClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

```
data = load_breast_cancer()
X = data.data
y = data.target

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
train_acc = []
test_acc = []
n_estimators_range = range(1, 51)

for n in n_estimators_range:
    model = AdaBoostClassifier(
        estimator=DecisionTreeClassifier(max_depth=1), # weak learner (stump)
        n_estimators=n,
        learning_rate=1,
        random_state=42
    )

    model.fit(X_train, y_train)

    # Predictions
    y_train_pred = model.predict(X_train)
    y_test_pred = model.predict(X_test)

    # Accuracy
    train_acc.append(accuracy_score(y_train, y_train_pred))
    test_acc.append(accuracy_score(y_test, y_test_pred))
```

```
plt.plot(n_estimators_range, train_acc, label='Train Accuracy')
plt.plot(n_estimators_range, test_acc, label='Test Accuracy')

plt.xlabel('Number of Weak Learners')
```

```
plt.ylabel('Accuracy')
plt.title('AdaBoost Performance Improvement')
plt.legend()
plt.grid()
plt.show()
```



